

APSC 2113 - Engineering Analysis I
Instructor: Professor Kartik Bulusu, MAE Dept., GWU
RQ1 - Fall 2026

Q1. Find the complete solution of the given initial value function

[15 pt]

$$\frac{dy}{dx} = 2y - 1 \quad y(1) = 1$$

Step 1: Particular Solution

Assume a constant solution $y_p = A$, so $\frac{dy_p}{dx} = 0$. Substituting gives $0 = 2A - 1$, which yields $A = \frac{1}{2}$. Thus, the particular solution is $y_p(x) = \frac{1}{2}$.

Step 2: Homogeneous Solution

Solve the homogeneous equation $\frac{dy_h}{dx} - 2y_h = 0$. Separating variables gives $\frac{dy_h}{y_h} = 2 dx$, which integrates to $\ln |y_h| = 2x + C$.

Exponentiating both sides yields the homogeneous solution $y_h(x) = Ce^{2x}$.

Step 3: Complete Solution

Combine the homogeneous and particular solutions using the superposition principle: $y(x) = y_h(x) + y_p(x)$.

This gives the general complete solution $y(x) = Ce^{2x} + \frac{1}{2}$.

Step 4: Applying Initial Condition

Apply $y(1) = 1$: $1 = Ce^{2(1)} + \frac{1}{2}$, which simplifies to $Ce^2 = \frac{1}{2}$, giving $C = \frac{1}{2}e^{-2}$. Substituting C back into the complete solution yields $y(x) = \frac{1}{2}e^{2(x-1)} + \frac{1}{2}$.

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Q1. Find the complete solution of the given initial value function

[15 pt]

$$\frac{dy}{dx} = 2y - 2 \quad y(1) = 1$$

Step 1: Particular Solution

Assume a constant solution $y_p = A$, so $\frac{dy_p}{dx} = 0$. Substituting gives $0 = 2A - 2$, which yields $A = 1$. Thus, the particular solution is $\mathbf{y_p(x) = 1}$.

Step 2: Homogeneous Solution

Solve the homogeneous equation $\frac{dy_h}{dx} - 2y_h = 0$. Separating variables gives $\frac{dy_h}{y_h} = 2 dx$, which integrates to $\ln |y_h| = 2x + C$.

Exponentiating both sides yields the homogeneous solution $\mathbf{y_h(x) = Ce^{2x}}$.

Step 3: Complete Solution

Combine the homogeneous and particular solutions: $y(x) = y_h(x) + y_p(x)$.

This gives the general complete solution $\mathbf{y(x) = Ce^{2x} + 1}$.

Step 4: Applying Initial Condition

Apply $y(1) = 1$: $1 = Ce^{2(1)} + 1$, which simplifies to $Ce^2 = 0$, giving $C = 0$.

Substituting $C = 0$ back into the complete solution yields $\mathbf{y(x) = 1}$.

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Q1. Find the complete solution of the given initial value function

[15 pt]

$$\frac{dy}{dx} = 2y - 3 \quad y(1) = 1$$

Step 1: Particular Solution

Assume a constant solution $y_p = A$, so $\frac{dy_p}{dx} = 0$. Substituting gives $0 = 2A - 3$, which yields $A = \frac{3}{2}$. Thus, the particular solution is $\mathbf{y_p(x) = \frac{3}{2}}$.

Step 2: Homogeneous Solution

Solve the homogeneous equation $\frac{dy_h}{dx} - 2y_h = 0$. Separating variables gives $\frac{dy_h}{y_h} = 2 dx$, which integrates to $\ln |y_h| = 2x + C$.

Exponentiating both sides yields the homogeneous solution $\mathbf{y_h(x) = Ce^{2x}}$.

Step 3: Complete Solution

Combine the homogeneous and particular solutions: $y(x) = y_h(x) + y_p(x)$.

This gives the general complete solution $\mathbf{y(x) = Ce^{2x} + \frac{3}{2}}$.

Step 4: Applying Initial Condition

Apply $y(1) = 1$: $1 = Ce^{2(1)} + \frac{3}{2}$, which simplifies to $Ce^2 = -\frac{1}{2}$, giving $C = -\frac{1}{2}e^{-2}$.

Substituting C back into the complete solution yields $\mathbf{y(x) = -\frac{1}{2}e^{2(x-1)} + \frac{3}{2}}$.

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Q1. Find the complete solution of the given initial value function

[15 pt]

$$\frac{dy}{dx} = 2y - 4 \quad y(1) = 1$$

Step 1: Particular Solution

Assume a constant solution $y_p = A$, so $\frac{dy_p}{dx} = 0$. Substituting gives $0 = 2A - 4$, which yields $A = 2$. Thus, the particular solution is $\mathbf{y_p(x) = 2}$.

Step 2: Homogeneous Solution

Solve the homogeneous equation $\frac{dy_h}{dx} - 2y_h = 0$. Separating variables gives $\frac{dy_h}{y_h} = 2 dx$, which integrates to $\ln |y_h| = 2x + C$.

Exponentiating both sides yields the homogeneous solution $\mathbf{y_h(x) = Ce^{2x}}$.

Step 3: Complete Solution

Combine the homogeneous and particular solutions: $y(x) = y_h(x) + y_p(x)$.

This gives the general complete solution $\mathbf{y(x) = Ce^{2x} + 2}$.

Step 4: Applying Initial Condition

Apply $y(1) = 1$: $1 = Ce^{2(1)} + 2$, which simplifies to $Ce^2 = -1$, giving $C = -e^{-2}$.

Substituting C back into the complete solution yields $\mathbf{y(x) = -e^{2(x-1)} + 2}$.
